

Averages

COMPLETE LEARNING GUIDE

What You'll Learn

Averages appear constantly across placement tests, both as standalone questions and as a hidden step inside age, speed, and data-interpretation problems, yet a large share of errors come from a single careless habit: updating the sum of a group without correspondingly updating the count (or vice versa) when a value is added, removed, or replaced. Averages questions reward a specific mental model — $\text{total} = \text{average} \times \text{count}$ — used forwards and backwards depending on what's given. This sub-module covers the basic sum-total relationship, averages of consecutive numbers and arithmetic progressions, the effect of adding or removing a value, the classic 'replace one group member' problem type, weighted averages across groups of different sizes, combining two groups' averages, and age-based average traps — each demonstrated with a complete numerical solution so the total-count bookkeeping becomes automatic.

1. Average Basics: The Sum-Total Relationship

$\text{Average} = \text{Sum of values} \div \text{Number of values}$, which rearranges to $\text{Sum} = \text{Average} \times \text{Count}$ — this rearranged form is what actually gets used in most exam questions, since the sum is rarely given directly. Every averages problem is really a bookkeeping exercise in tracking total and count correctly as they change.

- Find the average of 12, 15, 9, 22, and 18. $\text{Sum} = 12+15+9+22+18 = 76$. $\text{Average} = 76/5 = 15.2$.

2. Average of Consecutive Numbers and Arithmetic Progressions

For any evenly spaced (arithmetic) sequence — consecutive integers, consecutive even/odd numbers, or a general AP — the average is simply $(\text{first term} + \text{last term})/2$, since the values are symmetric around the midpoint. This avoids manually summing every term.

- Find the average of all integers from 21 to 35 (15 numbers). $\text{Average} = (21+35)/2 = 28$. $\text{Sum} = 28 \times 15 = 420$ (much faster than adding 15 numbers individually).

3. Effect on Average When a Value Is Added or Removed

Adding a new value changes both the sum (add the new value) and the count (add 1) — the new average is $(\text{old sum} + \text{new value})/(\text{old count} + 1)$. Removing a value subtracts it from the sum and subtracts 1 from the count. A very common error is updating only the sum and forgetting the count also changes.

- 8 numbers average 20 (sum = 160). A 9th number, 38, is added. New average = $(160+38)/9 = 198/9 = 22$.
- 10 numbers average 25 (sum = 250). The number 40 is removed. New average of the remaining 9 = $(250 - 40)/9 = 210/9 = 23.33$.

4. Replacing a Group Member: Change in Average

When one member of a fixed-size group is replaced by another, the TOTAL change in sum equals $(\text{change in average}) \times (\text{group size})$, and this total change equals exactly the difference between the new member's value and the old member's value. This gives a direct way to find the new (or old) member's value without recomputing every value in the group.

- The average weight of 20 boys is 45 kg. One boy weighing 35 kg leaves and is replaced by a new boy, after which the average becomes 45.5 kg. Total increase in sum = $20 \times 0.5 = 10$ kg. Since this increase equals $(\text{new boy's weight} - 35)$, the new boy weighs $35 + 10 = 45$ kg.

5. Weighted Average of Two or More Groups

When combining groups of different sizes, the combined average is NOT the simple average of the group averages — it must be weighted by each group's size: $\text{combined average} = (n_1 \cdot a_1 + n_2 \cdot a_2) / (n_1 + n_2)$, where n is each group's count and a is its average.

- Class A has 30 students averaging 70 marks; Class B has 20 students averaging 80 marks. Combined average = $(30 \times 70 + 20 \times 80) / (30 + 20) = (2100 + 1600) / 50 = 3700 / 50 = 74$ — closer to 70 than to the midpoint 75, because the larger group (A) pulls the weighted average toward itself.

6. Age-Based Average Problems

Age-average problems usually hinge on the same replace-or-add logic as weight/marks problems, but with an added twist: including a new member (like a teacher joining a class average) changes both the count and the sum, and the increase in average must be multiplied by the NEW total count, not the old one, to find the new member's value.

- The average age of 30 students in a class is 12 years. When the teacher's age is also included, the average increases by 1 year. New count = 31, new average = 13, so new total = $31 \times 13 = 403$. Original total = $30 \times 12 = 360$. Teacher's age = $403 - 360 = 43$ years.

7. Combining Averages Across Categories

When a question gives multiple sub-groups' individual averages and asks for an overall figure, first convert every average back into a total (using $\text{average} \times \text{count}$), sum all the totals, sum all the counts, and only then divide — never average the averages directly unless every group is exactly the same size.

- Three sections of a class have 25, 30, and 20 students, with average marks 60, 70, and 65 respectively. Overall average = $(25 \times 60 + 30 \times 70 + 20 \times 65) / (25 + 30 + 20) = (1500 + 2100 + 1300) / 75 = 4900 / 75 \approx 65.33$ — not the same as the simple average of 60, 70, 65, which would be 65.

Common Pitfalls

- Updating only the sum (not the count) or only the count (not the sum) when a value is added to or removed from a group.
- Averaging two or more group averages directly instead of weighting by each group's size.
- In replacement problems, computing the TOTAL change correctly but forgetting to add it to the old value rather than the average.
- Multiplying an average's increase by the OLD count instead of the NEW (post-inclusion) count when a new member joins the group.
- Assuming average of consecutive numbers requires summing every term instead of using $(\text{first} + \text{last}) / 2$.
- Treating 'divided among' or 'shared equally' language as an averages problem when it is actually a ratio problem, or vice versa.
- Forgetting that removing the highest or lowest value from a data set changes both the sum and the count, not just the sum.

Key Takeaways

- Sum = Average $\times$ Count — the relationship used forwards or backwards depending on what the question gives.
- Average of consecutive/evenly spaced numbers = $(\text{first term} + \text{last term}) / 2$.
- Adding or removing a value changes BOTH sum and count — update both before recomputing the average.
- Replacing a group member: $(\text{new value} - \text{old value}) = (\text{change in average}) \times (\text{group size})$.
- Weighted average = $(n_1 \cdot a_1 + n_2 \cdot a_2 + \dots) / (n_1 + n_2 + \dots)$; never average averages of unequal-sized groups directly.
- When a new member joins and the average shifts by d , the new member's value = old average + $d \times \text{new total count}$.
- To combine several sub-group averages into one overall average, convert each back to a total first, then divide by the combined count.

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